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Zeng et al.

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(54) **TRANSPERINEAL PUNCTURE GUIDING BRACKET**

(58) **Field of Classification Search**

None

See application file for complete search history.

(71) Applicant: **JingFang Precision Medical Device (Shenzhen) Co., Ltd.**, Guangdong (CN)

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(72) Inventors: **Junhua Zeng**, Guangdong (CN); **Yi Han**, Guangdong (CN)

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(73) Assignee: **JingFang Precision Medical Device (Shenzhen) Co., Ltd.**, Shenzhen (CN)

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ABSTRACT

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A transperineal puncture guiding bracket is provided, comprising a probe fixing assembly and a guiding assembly, wherein a mounting plate is set on top of the probe fixing assembly, the guiding assembly comprises a needle grooved plate and a cover plate, and the needle grooved plate is detachably combined to a first side surface of the mounting plate by a first vertical insertion structure, a first hanging structure and a first buckling structure, the cover plate is detachably combined with the needle grooved plate by a second vertical insertion structure, a second hanging structure and a second buckling structure, thereby the cover plate is detachably attached to the needle grooved plate to form a plurality of needle guiding holes. In the bracket, needled grooved plates of different specifications can be replaced according to the puncture needles, and the puncture needle can be easily separated from the bracket.

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A61B 17/34 (2006.01)

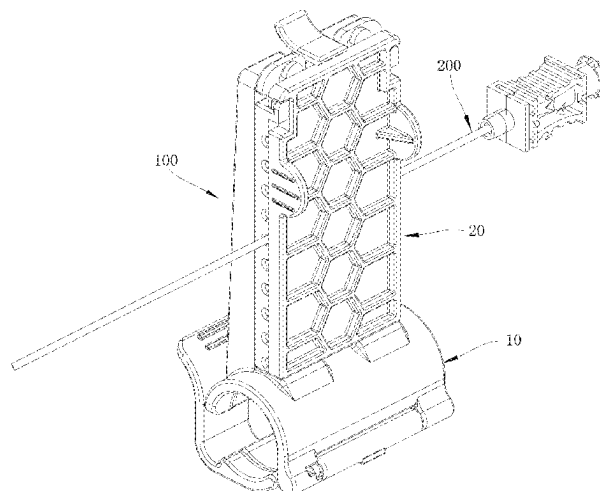
A61B 8/00 (2006.01)

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CPC **A61B 17/3403** (2013.01); **A61B 8/12** (2013.01); **A61B 8/4455** (2013.01); **A61B 2017/3411** (2013.01); **A61B 2017/3413** (2013.01)

10 Claims, 10 Drawing Sheets



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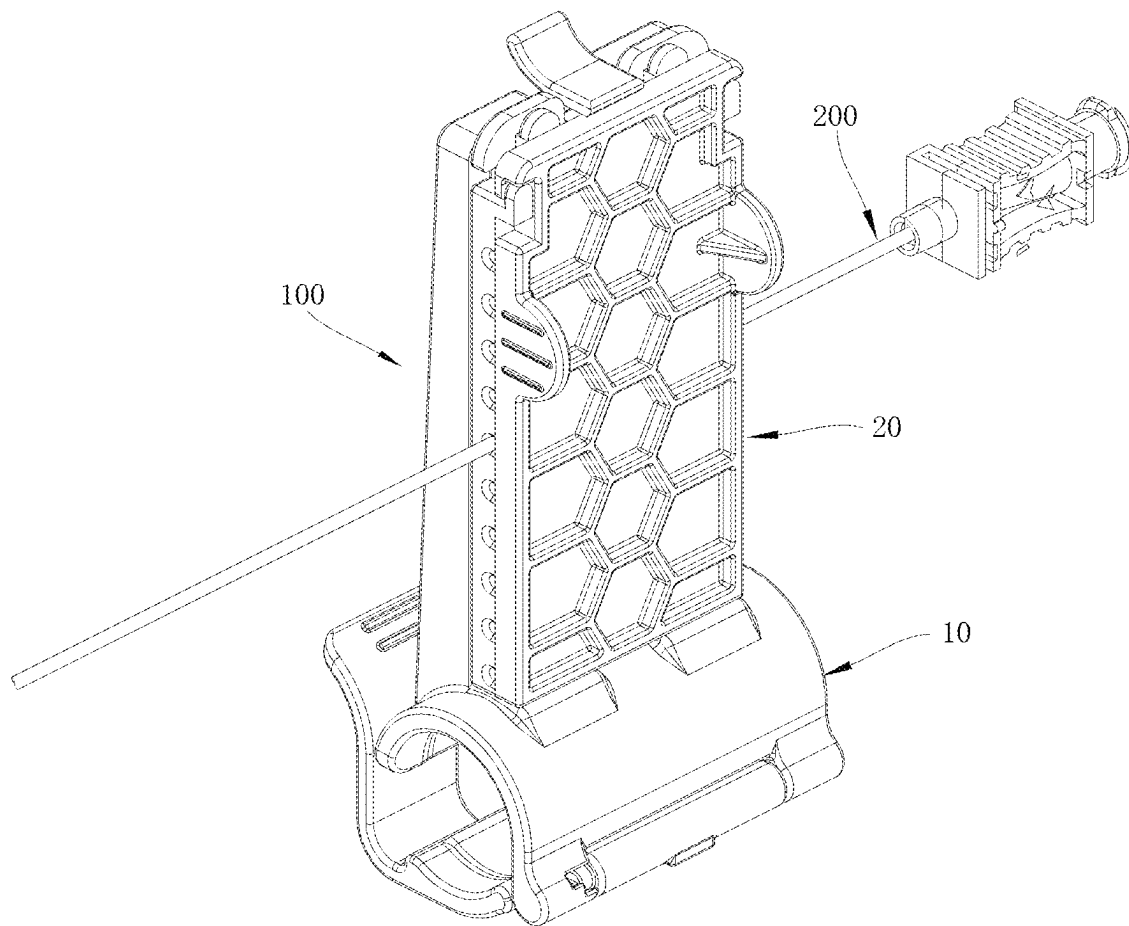


FIG. 1

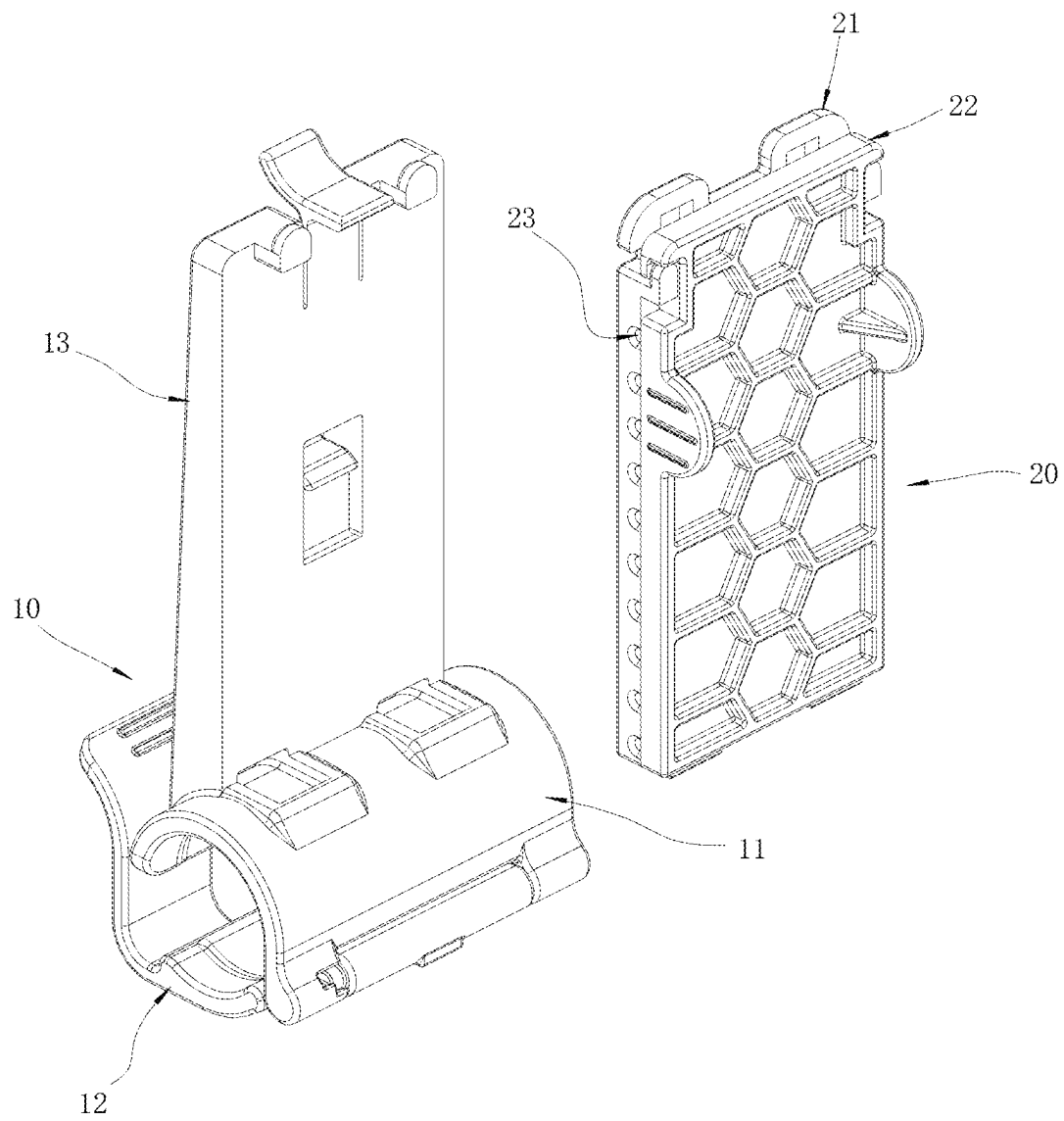


FIG. 2

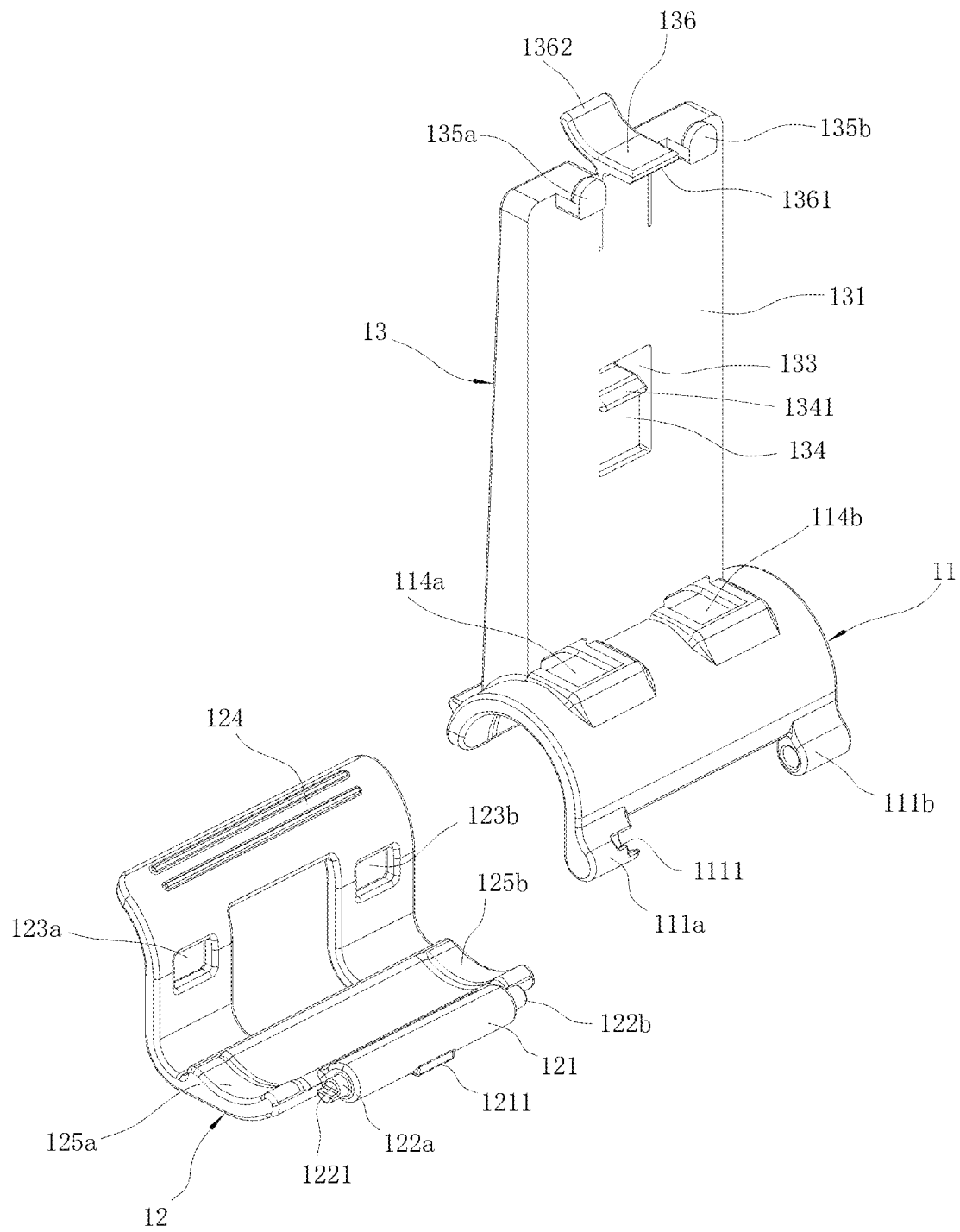


FIG. 3

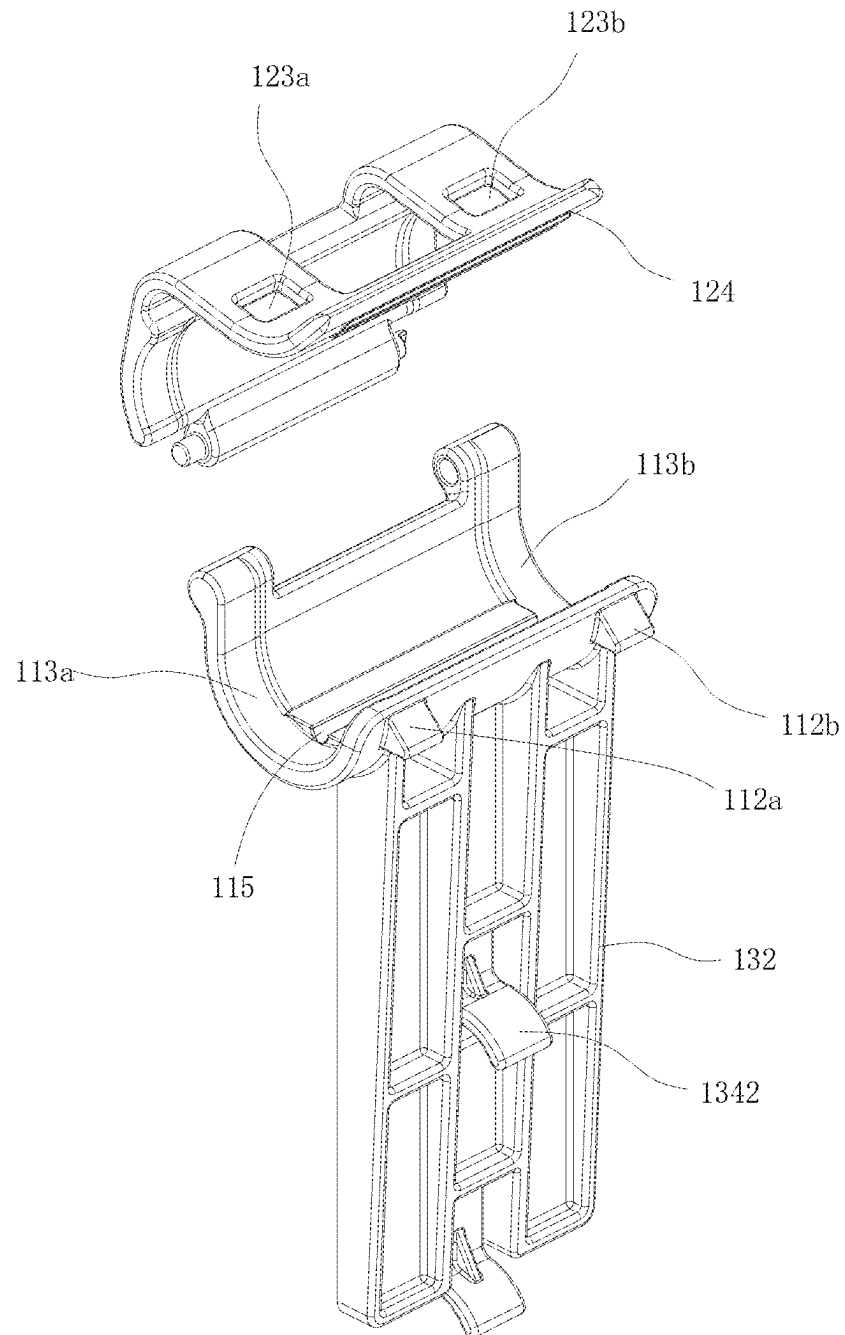


FIG. 4

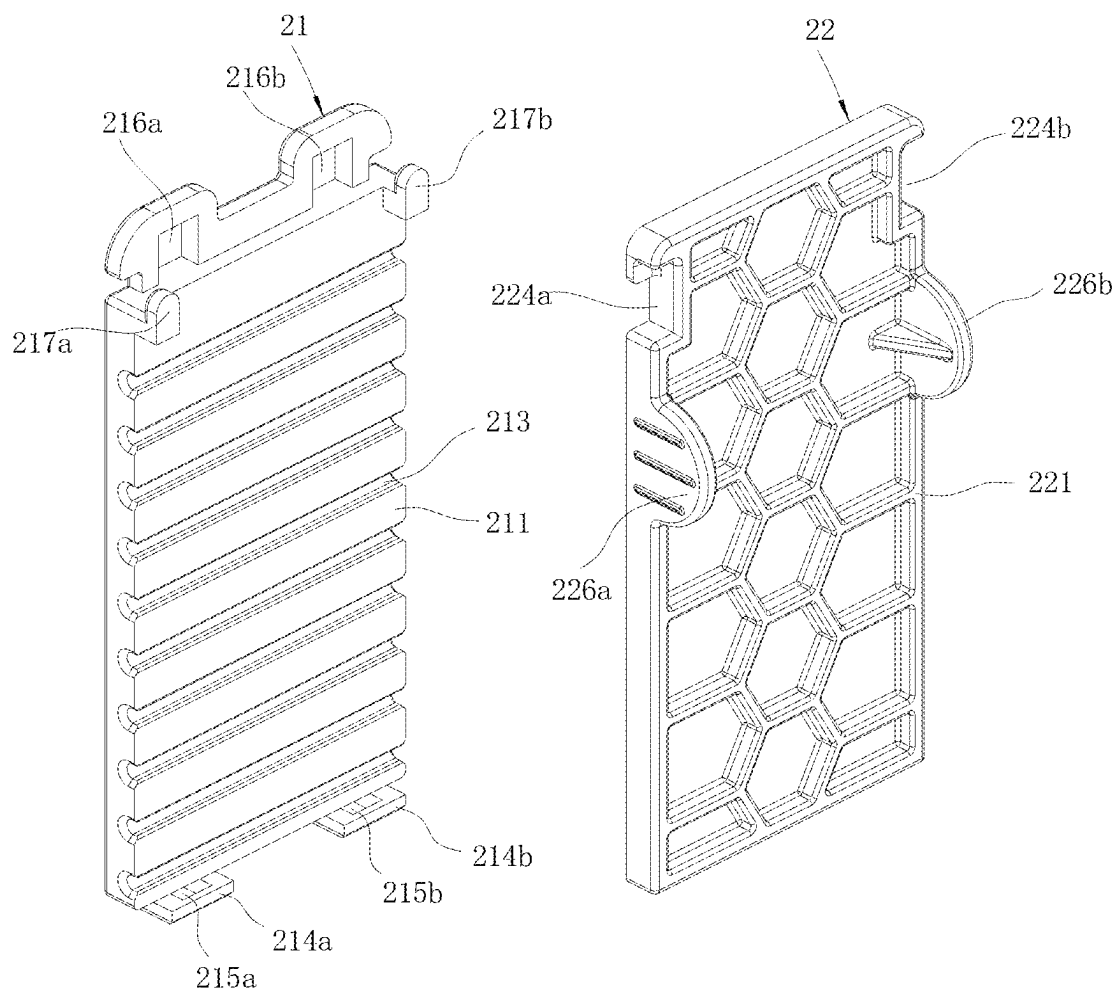


FIG. 5

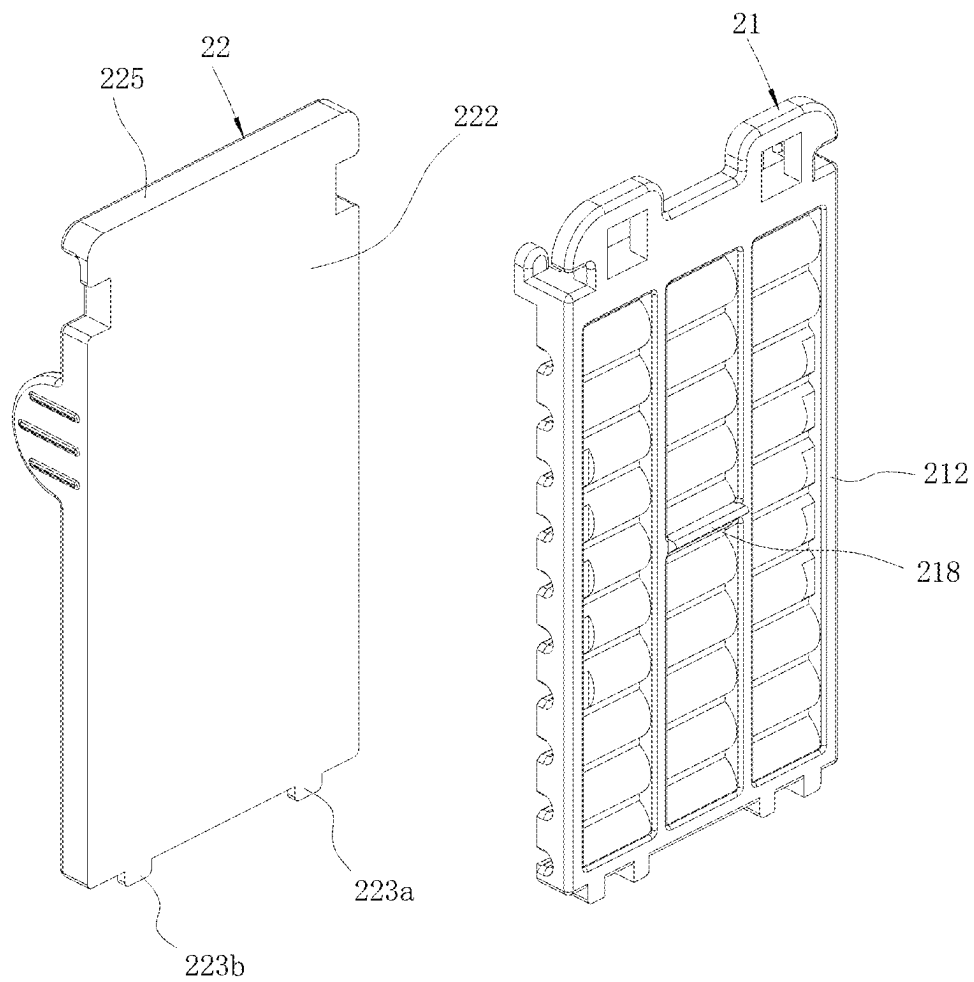


FIG. 6

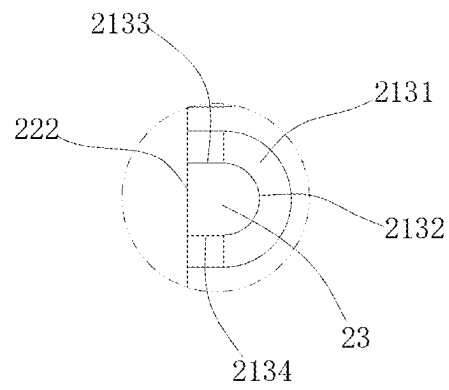


FIG. 7

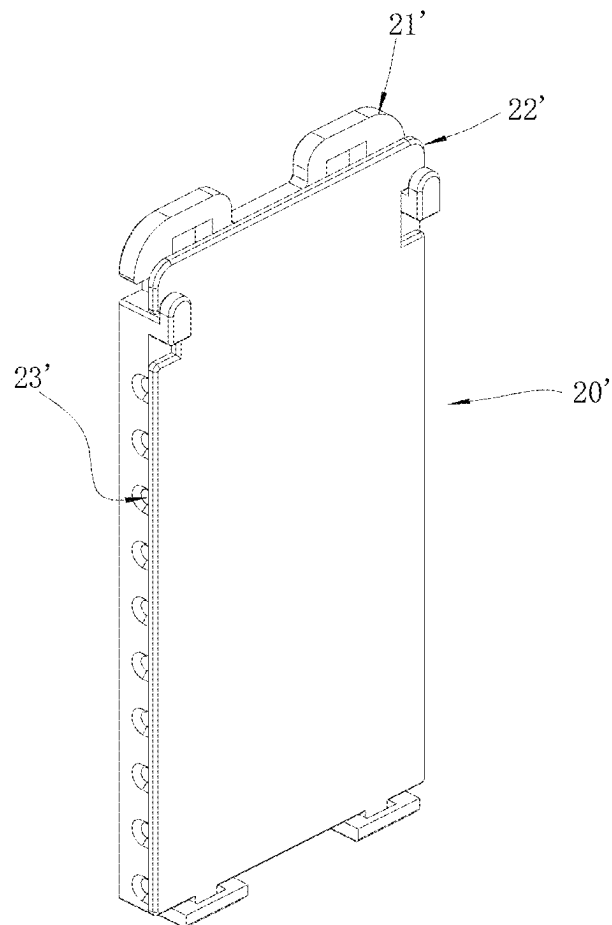


FIG. 8

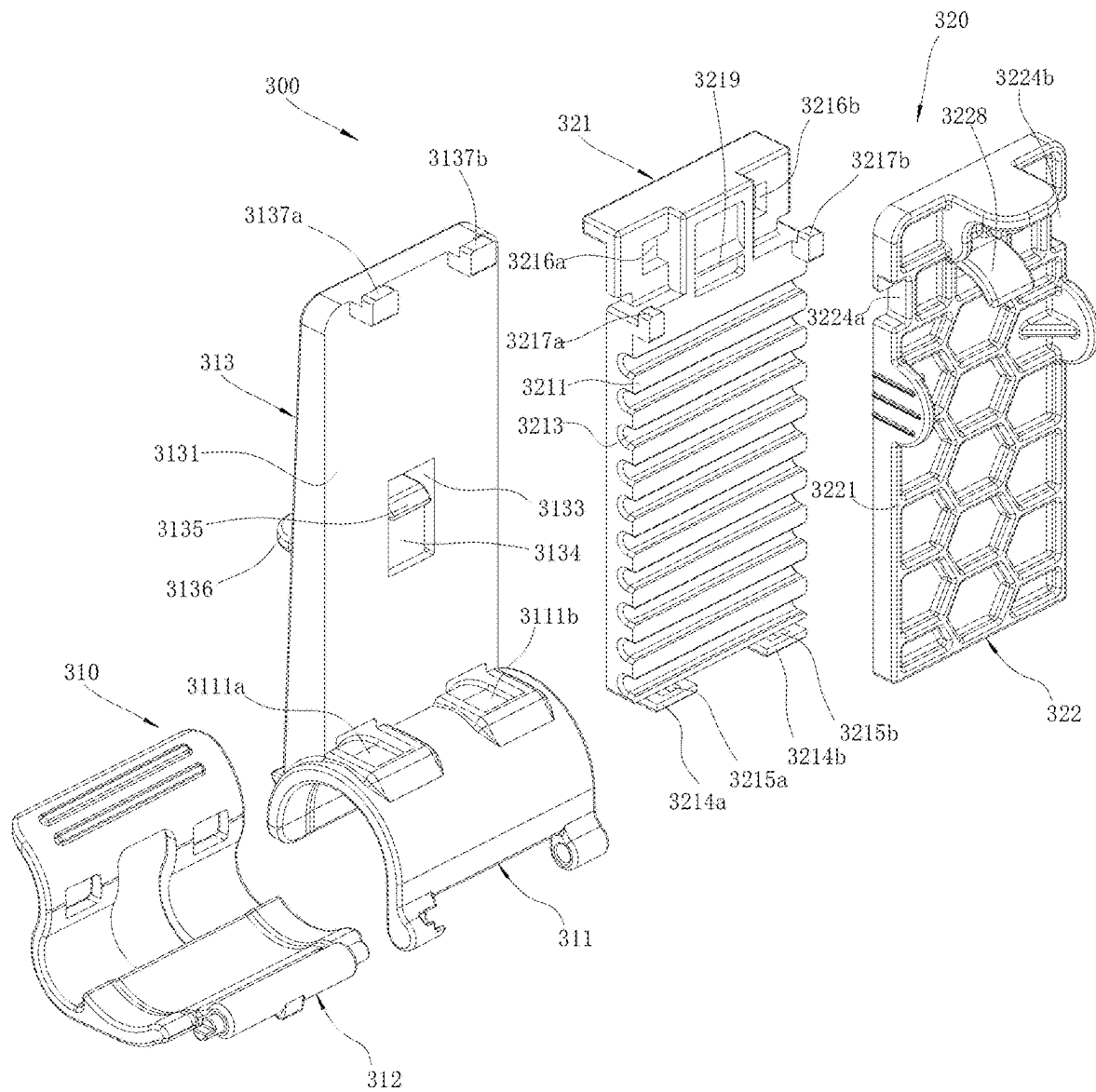


FIG. 9

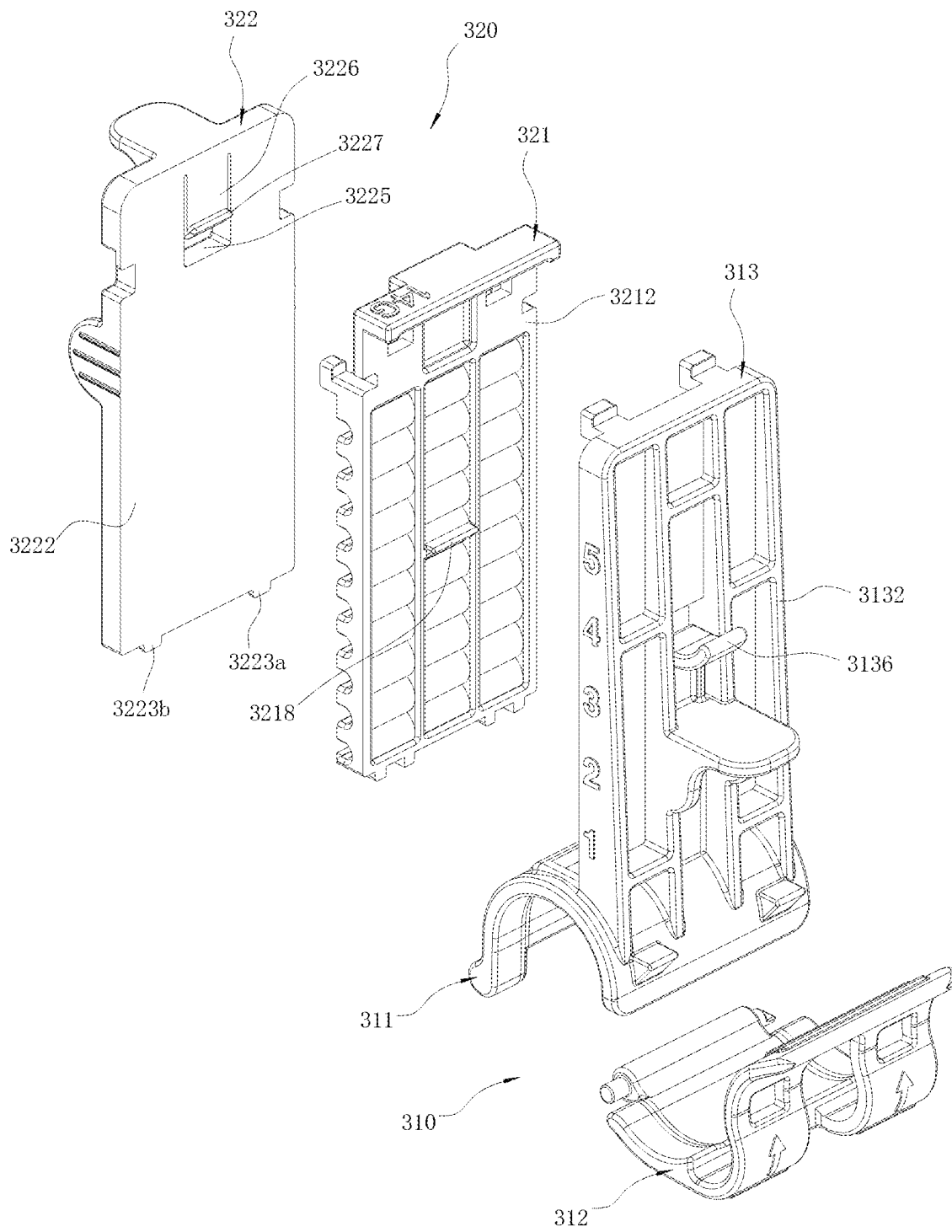


FIG. 10

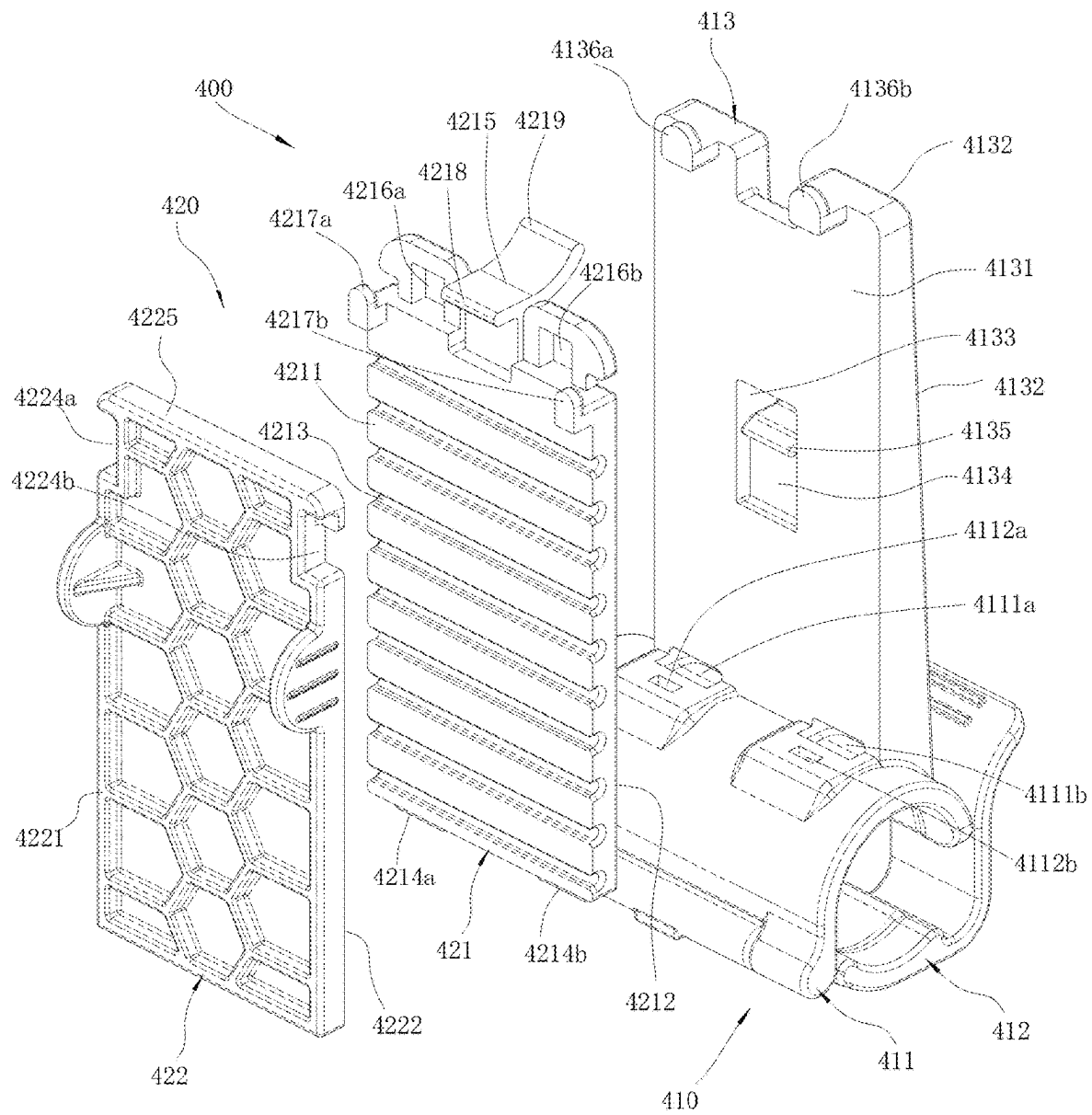


FIG. 11

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TRANSPERINEAL PUNCTURE GUIDING BRACKET

FIELD

The present application generally relates to ultrasound intervention medical auxiliary devices, more particularly to a transperineal puncture guiding bracket.

BACKGROUND

With changes of people's lifestyle in recent years, an incidence rate of prostate cancer in China has increased year by year, and it has ranked first in malignant tumors of the urinary system (male). Early detection and timely treatment can save lives of patients. Real time ultrasound guided prostate biopsy is a best choice for diagnosing prostate cancers due to its small trauma and high accuracy. There are two ways to perform prostate biopsy surgery: one is through the perineum, and the other is through the rectum. During transrectal puncture biopsy surgery, a biopsy needle is inserted into the prostate site through an entrance of the anus and intestines, and the rectal mucosa is prone to bleeding and infection after surgery. During transperineal puncture biopsy surgery, a biopsy needle is inserted into the prostate site through the external perineum, thus the surgery can be aseptic, has a low risk of infection, comprehensive sampling, and higher safety. Transperineal puncture biopsy is more conducive to detecting early tumors and reducing postoperative adverse reactions, and is currently a main-stream prostate biopsy method.

In the process of transperineal prostate biopsy surgery, an intracavitary ultrasound probe is equipped with a puncture guiding bracket for use. A front end of the puncture guiding bracket needs to be close to a patient's perineum, and a biopsy needle needs to pass through a guide hole on the puncture guiding bracket to puncture into the prostate gland. Commonly used puncture guiding brackets currently have the following limitations: 1. Due to use of a closed needle hole structure for guiding a biopsy needle, the puncture guiding bracket cannot be separated from the biopsy needle, which limits the flexible operation of doctors. 2. A needle grooved plate and a bracket body are integrated to form a whole structure, thus different puncture guiding brackets are required when using biopsy needles of different specifications, which brings inconvenience to doctors during surgical operations.

SUMMARY

The technical problem to be solved by the present application is to provide a transperineal puncture guiding bracket that can replace needle grooved plates of different specification according to puncture needles of different specifications and facilitate separation of puncture needles in view of the above-mentioned drawbacks of the prior art.

The technical solution adopted by the present application to solve the technical problem thereof is to provide a transperineal puncture guiding bracket, comprising a probe fixing assembly and a guiding assembly, wherein the probe fixing assembly is fixedly installed with an intracavitary ultrasound probe, the guiding assembly provides a plurality of vertically distributed and parallel needle guiding holes along a puncture direction, wherein, a mounting plate is set on top of the probe fixing assembly, and the guiding assembly further comprises a needle grooved plate and a cover plate, the needle grooved plate has a first side surface and a

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second side surface opposite to the first side surface, with the first side surface of the needle grooved plate being provided with a plurality of vertically distributed and parallel needle guiding grooves; and a lower portion of the needle grooved plate is detachably combined with the top of the probe fixing assembly adjacent to the mounting plate by a first vertical insertion structure, an upper portion of the needle grooved plate is detachably hung with an upper portion of the mounting plate by a first hanging structure, and the second side surface of the needle grooved plate is detachably buckled with a first side surface of the mounting plate by a first buckling structure, thereby the second side surface of the needle grooved plate is detachably attached to the first side surface of the mounting plate; and a lower portion of the cover plate is detachably combined with the lower portion of the needle grooved plate or the top of the probe fixing assembly by a second vertical insertion structure, an upper portion of the cover plate is detachably hung with the upper portion of the needle grooved plate by a second hanging structure, and the cover plate is further detachably buckled with the upper portion of the mounting plate, the upper portion of the needle grooved plate or the top of the probe fixing assembly through a second buckling structure, thereby a second side surface of the cover plate is detachably attached to the first side surface of the needle grooved plate to close the plurality of needle guiding grooves to form the plurality of needle guiding holes.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the first vertical insertion structure comprises first limiting feet provided at the lower portion of the needle grooved plate which extend out from the first side surface of the needle grooved plate and limiting grooves formed at the top of the probe fixing assembly for insertion of the first limiting feet; and the second vertical insertion structure comprises second limiting feet set at the lower portion of the cover plate and limiting holes on the first limiting feet of the needle grooved plate for insertion of the second limiting feet.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the first vertical insertion structure comprises first limiting feet provided at the lower portion of the needle grooved plate and first limiting grooves set at the top of the probe fixing assembly for insertion of the first limiting feet; and the second vertical insertion structure comprises second limiting feet set at the lower portion of the cover plate and second limiting grooves formed at the top of the probe fixing assembly for insertion of the second limiting feet.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the first hanging structure comprises first hanging holes at the upper portion of the needle grooved plate and first hooks at the upper portion of the mounting plate which extend out from the first side surface of the mounting plate to hang into the first hanging holes; and the second hanging structure comprises second hanging holes at the upper portion of the cover plate and second hooks at the upper portion which extend out from the first side surface of the needle grooved plate to hang into the second hanging holes.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the first buckling structure comprises a first limiting buckle set in the middle of the second side surface of the needle grooved plate and a first elastic buckle set in the middle of the mounting plate, wherein a through-hole is set in the middle of the mounting plate that connects the first side surface and a second side surface of the mounting plate, and the first elastic buckle is

set inside the through-hole, the first elastic buckle has a first buckling position that is engaged with the first limiting buckle and a first operating part that extends out of the second side surface of the mounting plate for releasing engagement between the first buckling position and the first limiting buckle.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the second buckling structure comprises a second elastic buckle set at the upper portion of the needle grooved plate, wherein the second elastic buckle has a second buckling position for pressing a top surface of the cover plate and a second operating part for releasing the second buckling position from pressing the top surface of the cover plate.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the second buckling structure comprises a second elastic buckle set at the upper portion of the mounting plate, wherein the second elastic buckle has a second buckling position for pressing a top surface of the cover plate and a second operating part for releasing the second buckling position from pressing the top surface of the cover plate.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the second buckling structure comprises a second limiting buckle set at the upper portion of the first side surface of the needle grooved plate and a second elastic buckle at the corresponding upper portion of the cover plate, wherein a through-hole is set at the upper portion of the cover plate that connects a first side surface and the second side surface of the cover plate, and the second elastic buckle is set inside the through-hole, the second elastic buckle has a second buckling position that is engaged with the second limiting buckle and a second operating part that extends out of the first side surface of the cover plate for releasing engagement between the second buckling position and the second limiting buckle.

In the transperineal puncture guiding bracket according to an embodiment of the present application, the probe fixing assembly comprises an upper fixing member and a lower fixing member, wherein the upper fixing member and the lower fixing member work together to hold outer surface of the intracavitary ultrasound probe, and the mounting plate is set on top of the upper fixing member.

In the transperineal puncture guiding bracket according to an embodiment of the present application, upper wrapping surfaces are formed on a lower surface of the upper fixing member for fitting upper surface of the intracavitary ultrasound probe, and lower wrapping surfaces are formed on an upper surface of the lower fixing member for fitting lower surface of the intracavitary ultrasound probe, and the upper fixing member and the lower fixing member are rotationally connected on one side and buckled on the opposite side to tightly hold the intracavitary ultrasound probe through the upper wrapping surfaces and the lower wrapping surfaces.

Implementing the transperineal puncture guiding bracket of the present application has the following beneficial effects: The transperineal puncture guiding bracket according to the embodiments of the present application can guide puncture needles of different specifications by replacing the guiding assembly with needle holes of different specifications, without a need of replacing the entire transperineal puncture guiding bracket, which facilitates surgical operations of doctors. With the transperineal puncture guiding bracket according to the embodiments of the present application, the puncture needle can be easily separated from the transperineal puncture guiding bracket during surgery by

separating the cover plate from the needle grooved plate of the guiding assembly, making the doctor's operation more flexible.

BRIEF DESCRIPTION OF THE DRAWINGS

The present application will be further described with reference to the accompanying drawings and embodiments, in which:

FIG. 1 is a schematic structural diagram of a transperineal puncture guiding bracket according to a first embodiment of the present application;

FIG. 2 is an exploded structural diagram of the transperineal puncture guiding bracket shown in FIG. 1;

FIG. 3 is an exploded structural diagram of a probe fixing assembly shown in FIG. 2;

FIG. 4 is an exploded structural diagram of the probe fixing assembly shown in FIG. 2 from another perspective;

FIG. 5 is an exploded structural diagram of a guiding assembly shown in FIG. 2;

FIG. 6 is an exploded structural diagram of the guiding assembly shown in FIG. 2 from another perspective;

FIG. 7 is a planar structural diagram of a needle inlet end of a needle guiding hole according to the first embodiment of the present application;

FIG. 8 is a schematic structural diagram of a guiding assembly according to a second embodiment of the present application;

FIG. 9 is an exploded structural diagram of a transperineal puncture guiding bracket according to a third embodiment of the present application;

FIG. 10 is an exploded structural diagram of the transperineal puncture guiding bracket shown in FIG. 9 from another perspective;

FIG. 11 is schematic structural diagram of a transperineal puncture guiding bracket according to a fourth embodiment of the present application.

DETAILED DESCRIPTION

To explain objects, technical solutions and advantages of the present application more clearly, the present application will be further described with reference to the accompanying drawings and embodiments in the following. It should be understood that, the specific embodiments described here are only for explanation, but not for limitation to the present application. And, the embodiments and features in the embodiments of the present application can be combined with each other without conflict.

FIG. 1 shows a schematic structural diagram of a transperineal puncture guiding bracket 100 guiding a puncture needle 200 according to a first embodiment of the present application, and FIG. 2 shows an exploded structural diagram of the transperineal puncture guiding bracket 100. As shown in FIG. 1 and FIG. 2, the transperineal puncture guiding bracket 100 comprises a probe fixing assembly 10 and a guiding assembly 20. The probe fixing assembly 10 is used for fixed installation with an intracavitary ultrasound probe. The guiding assembly 20 is detachably installed on the probe fixing assembly 10. The guiding assembly 20 provides a plurality of needle guiding holes 23 along a puncture direction, which are vertically equidistant and parallel to each other, used to guide the puncture needle 200 to pass through a needle guiding hole 23 for biopsy puncture. The probe fixing assembly 10 further comprises an upper fixing member 11, a lower fixing member 12, and a mounting plate 13. The upper fixing member 11 and the

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lower fixing member 12 work together to hold outer surface of the intracavitary ultrasound probe for fixation. The mounting plate 13 is set on the upper fixing member 11 for detachable installation of the guiding assembly 20. The guiding assembly 20 further comprises a needle grooved plate 21 and a cover plate 22, which are also detachably combined to form the plurality of vertically equidistant and parallel needle guiding holes 23. The transperineal puncture guiding bracket 100 according to the above embodiment of the present application can guide puncture needles 200 of different specifications by replacing the guiding assembly 20 with needle holes 23 of different specifications, without a need of replacing the entire transperineal puncture guiding bracket 100, which facilitates surgical operations of doctors. With the transperineal puncture guiding bracket 100 according to the above embodiment of the present application, the puncture needle 200 can be easily separated from the transperineal puncture guiding bracket 100 during surgery by separating the cover plate 22 from the needle grooved plate 21 of the guiding assembly 20, making the doctor's operation more flexible.

FIG. 3 and FIG. 4 respectively show exploded structural diagrams of the probe fixing assembly 10 from different perspectives. As shown in FIG. 3 and FIG. 4, the probe fixing assembly 10 comprises the upper fixing member 11, the lower fixing member 12, and the mounting plate 13. The upper fixing member 11 has a roughly arc shaped outer contour that matches the outer surface of the intracavitary ultrasound probe. The mounting plate 13 is fixedly set on one side along the puncture direction on top of the upper fixing member 11, and can be integrated molded with the upper fixing member 11. A positioning groove 115 that matches a positioning convex bar on the intracavitary ultrasound probe is formed on a lower surface of the upper fixing member 11. Rotation of the upper fixing member 11 can be restricted by aligning and inserting the positioning groove 115 with the positioning convex bar on the intracavitary ultrasound probe. Upper wrapping surfaces 113a and 113b are formed on front and rear ends along the puncture direction of the lower surface of the upper fixing member 11 respectively, which are used for fitting the upper surface of the intracavitary ultrasound probe. In addition, there are shaft sleeves 111a and 111b at the front and rear ends on one side of the upper fixing member 11, and a notch 1111 is formed at an inlet end of one shaft sleeve 111a. On the other side opposite to the two shaft sleeves 111a and 111b of the upper fixing member 11, there are also equipped with two limiting buckles 112a and 112b that are spaced forward and backward. Correspondingly, the lower fixing member 12 also has a roughly arc shaped outer contour that matches the outer surface of the intracavitary ultrasound probe. Two lower wrapping surfaces 125a and 125b are formed at front and rear ends of the upper surface of the lower fixing member 12 respectively, which are fitted with the outer surface of the intracavitary ultrasound probe. In addition, a connecting rod 121 is formed on the lower fixing member 12 on one side corresponding to the two shaft sleeves 111a and 111b of the upper fixing member 11, and two rotating shafts 122a and 122b that rotationally fit with the two shaft sleeves 111a and 111b extend coaxially from both ends of the connecting rod 121. A semi-cylindrical protrusion 1221 is provided at the end of the rotating shaft 122a corresponding to the shaft sleeve 111a having a notch 1111. After the rotating shaft 122b of the lower fixing member 12 is aligned with and inserted into the shaft sleeve 111b of the upper fixing member 11, the lower fixing member 12 is rotated so as to align and insert the semi-cylindrical protrusion 1221 on

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the rotating shaft 122a into the notch 1111 on the shaft sleeve 111a of the upper fixing member 11, then the lower fixing member 12 is further rotated around the rotating shafts 122a and 122b till an indicator protrusion 1211 on the connecting rod 121 turns to the outside, thus the upper fixing member 11 and the lower fixing member 12 are rotationally connected on the one side (see FIG. 2). On the other side of the lower fixing member 12, a handheld unit 124 extends upwards from the lower fixing member 12, and two retaining rings 123a and 123b are formed on the handheld unit 124, to match the two limiting buckles 112a and 112b on the corresponding other side of the upper fixing member 11. After the upper wrapping surfaces 113a and 113b of the upper fixing member 11 are tightly attached to the upper surface of the intracavitary ultrasound probe with the positioning convex bar on the upper surface of the intracavitary ultrasound probe being aligned and embedded into the positioning groove 115, the lower fixing member 12 is rotated by the handheld unit 124 around axis of the rotating shafts 122a and 122b relative to the upper fixing member 11, till the lower wrapping surfaces 125a and 125b of the lower fixing member 12 are tightly attached to the lower surface of the intracavitary ultrasound probe. Then, the two retaining rings 123a and 123b are interlocked with the two limiting buckles 112a and 112b by force acting on the handheld unit 124 to make the handheld unit 124 generating elastic deformation adapting to the shape of the intracavitary ultrasound probe, thus the upper fixing member 11 and the lower fixing member are clasped on the corresponding other side to tightly hold the intracavitary ultrasound probe. By reverse force acting on the handheld unit 124, the two retaining rings 123a and 123b on the handheld unit 124 can be released from the two limiting buckles 112a and 112b, and then the intracavitary ultrasound probe can be separated by reversibly rotating the lower fixing member 12. Due to a certain deformation curvature of the handheld unit 124, it can withstand clamping forces of different deformation amounts, and therefore the probe fixing assembly 10 can be compatible with intracavitary ultrasound probes with different diameters within a certain range. Moreover, there is avoidance design that does not contact with the intracavitary ultrasound probe between the respective upper wrapping surfaces 113a and 113b of the upper fixing member 11 and the respective lower wrapping surfaces 125a and 125b of the lower fixing member 12. Thus warping of the probe fixing assembly can be avoided after installation on the intracavitary ultrasound probe.

Again as shown in FIG. 3 and FIG. 4, the mounting plate 13 is fixedly set on one side along the puncture direction on top of the upper fixing member 11, having a first side surface 131 and a second side surface 132 opposite to the first side surface 131. There is a through-hole 133 in the middle of the mounting plate 13 that connects the first side surface 131 and the second side surface 132, and a first elastic buckle 134 is set in the through-hole 133 for clamping the needle grooved plate 21 of the guiding assembly 20. The first elastic buckle 134 has a first buckling position 1341 extending out from the first side surface 131 and a first operating part 1342 extending out from the second side surface 132. A second elastic buckle 136 is set in the middle on top of the mounting plate 13, which is used for clamping the cover plate 22 of the guiding assembly 20. According to different embodiments of the present application, the second elastic buckle 136 can also be set at other positions on upper portion of the mounting plate 13, not limited to the top of the mounting plate 13 in the illustrated embodiment. The second elastic buckle 136 also has a second buckling position 1361 extend-

ing out from the first side surface **131** and a second operating part **1362** extending out from the second side surface **132**. The mounting plate **13** is also equipped with two first hooks **135a** and **135b** on its first side surface **131**, which extend out from the first side surface **131** at front and rear ends on the top of the first side surface **131** respectively, for hanging the needle grooved plate **21** of the guiding assembly **20**. Similarly, according to different embodiments of the present application, the first hooks **135a** and **135b** may also be provided at other positions on the upper portion of the mounting plate **13**, not limited to the top of the mounting plate **13** in the illustrated embodiment. There are also two limiting grooves **114a** and **114b** on the top of the upper fixing member **11** adjacent to the mounting plate **13**, which are used to install and position the needle grooved plate **21** of the guiding assembly **20** in cooperation with the mounting plate **13**. Regarding specific implementation of the detachable combination structure between the mounting plate **13** and the guiding assembly **20**, a more detailed description will be provided in conjunction with the description of the guiding assembly **20** in the below.

FIG. 5 and FIG. 6 show exploded structural diagrams of the guiding assembly **20** from different perspectives respectively. As shown in FIG. 5 and FIG. 6, the guiding assembly **20** is composed of a needle grooved plate **21** and a cover plate **22** that are detachably combined with each other. The needle grooved plate **21** has a first side surface **211** and a second side surface **212** opposite to the first side surface **211**. On the first side surface **211** of the needle grooved plate **21**, there are a plurality of needle guiding grooves **213** which are vertically equidistant and parallel to each other. The needle guiding grooves **213** pass through the front and rear ends of the needle grooved plate **21** along the puncture direction, and an insertion guiding surface **2131** is formed at least at the needle inlet end (as shown in FIG. 7) to facilitate needle insertion. The insertion guiding surface **2131** is preferably a guiding slope. The cover plate **22** has a first side surface **221** and a second side surface **222** opposite to the first side surface **221**. After the cover plate **22** is combined with the needle grooved plate **21**, the second side surface **222** of the cover plate **22** is tightly adhered to the first side surface **211** of the needle grooved plate **21**, closing open sides of the plurality of needle guiding grooves **213** to form the plurality of needle guiding holes **23**. As shown in FIG. 7, each of the needle guiding grooves **213** is preferably a U-shaped groove, which has a half-circle right limiting cambered surface **2132**, an upper limiting plat surface **2133**, and a lower limiting plat surface **2134** when viewed from the puncture direction, plus the second side surface **222** of the cover plate **22** serving as a left limiting plat surface to form a D-shaped needle guiding hole **23**, which limits position of the puncture needle **200** up, down, left, and right, so that the puncture needle **200** can only move axially.

As shown in FIG. 5 and FIG. 6 again, two first limiting feet **214a** and **214b** are provided at a lower portion (preferably at the bottom) of the needle grooved plate **21**, which extend out from the first side surface **211** at front and rear ends of the needle grooved plate **21** respectively, for insertion into the limiting grooves **114a** and **114b** on top of the upper fixing member **11**. There are further limiting holes **215a** and **215b** on the two first limiting feet **214a** and **214b** respectively, for insertion and installation of two second limiting feet **223a** and **223b** provided at a corresponding lower portion (preferably at the bottom) of the cover plate **22**. Two second hooks **217a** and **217b** are provided at an upper portion of the needle grooved plate **21**, which extend out from the first side surface **211** at the front and rear ends

of the needle grooved plate **21** respectively. Corresponding to the second hooks **217a** and **217b**, two second hanging holes **224a** and **224b** are formed at the front and rear ends of the cover plate **22**, which are used for hanging the second hooks **217a** and **217b**. After the second hanging holes **224a** and **224b** of the cover plate **22** are aligned and hung with the second hooks **217a** and **217b** of the needle grooved plate **21**, the second limiting feet **223a** and **223b** at the bottom of the cover plate **22** are inserted into the limiting holes **215a** and **215b** on the first limiting feet **214a** and **214b** at the bottom of the needle grooved plate **21** by moving downwards the cover plate **22** with its second side surface **222** closely contacting the first side surface **211** of the needle grooved plate **21**, thus the cover plate **22** is combined with the needle grooved plate **21**. Reverse operation can also separate the cover plate **22** from the needle grooved plate **21**. In addition, two first hanging holes **216a** and **216b** spaced back and forth are formed at the upper portion of the needle grooved plate **21**, which pass through the first side surface **211** to the second side surface **212** of the needle grooved plate **21**. These two first hanging holes **216a** and **216b** are matched with the two first hooks **135a** and **135b** at the top of the mounting plate **13**, for the two first hooks **135a** and **135b** to be hung in. A first limiting buckle **218** is also formed in the middle of the second side surface **212** of the needle grooved plate **21**, which is used to engage with the first elastic buckle **134** of the mounting plate **13**. On the first side surface **221** of the cover plate **22**, there are also two handheld members **226a** and **226b** extending out from the first side surface **221** at the front and rear ends of the cover plate **22**, which are convenient for operation. A process of installing the guiding assembly **20** composed of the combined needle grooved plate **21** and the cover plate **22** onto the probe fixing assembly **10** is as follows: pressing the second operating part **1362** of the second elastic buckle **136** at the top of the mounting plate **13** to lift the second buckling position **1361** of the second elastic buckle **136**; then, aligning and hanging the first hanging holes **216a** and **216b** on the upper portion of the needle grooved plate **21** with the first hooks **135a** and **135b** on the top of the mounting plate **13**, and moving downwards the needle grooved plate **21** with the second side surface **212** of the needle grooved plate **21** being closely attached to the first side surface **131** of the mounting plate **13**, and inserting the first limiting feet **214a** and **214b** on the lower portion of the needle grooved plate **21** into the limiting grooves **114a** and **114b** on top of the upper fixing member **11**, and engaging the first limiting buckle **218** in the middle of the second side surface **212** of the needle grooved plate **21** with the first elastic buckle **1341** in the middle of the mounting plate **13**; and then, release the pressing on the second operating part **1362** of the second elastic buckle **136** at the top of the mounting plate **13** to make the second buckling position **1361** of the second elastic buckle **136** press against a top surface **225** of the cover plate **22**. Thus the guiding assembly **20** is installed and positioned on the probe fixing assembly **10**. Except that the needle grooved plate **21** is firstly combined with the cover plate **22** and then is installed onto the probe fixing assembly **10** as described above, the transperineal puncture guiding bracket **100** according to the embodiments of the present application may also be configured with the needle grooved plate **21** being firstly assembled with the mounting plate **13** of the probe fixing assembly **10** and then being combined with the cover plate **22**.

According to the above embodiment of the present application, the lower portion of the needle grooved plate **21** is detachably combined with the upper fixing member **11** by a

first vertical insertion structure composed of the first limiting feet **214a** and **214b** and the limiting grooves **114a** and **114b**, the upper portion of the needle grooved plate **21** is detachably hung with the upper portion of the mounting plate **13** by a first hanging structure composed of the first hanging hole **216a** and **216b** and the first hooks **135a** and **135b**, and the middle portion of the second side surface **212** of the needle grooved plate **21** is detachably buckled with the middle portion of the first side surface **131** of the mounting plate **13** by a first buckling structure composed of the first limiting buckle **218** and the first elastic buckle **134**, thereby the second side surface **212** of the needle grooved plate **21** is detachably attached to the first side surface **131** of the mounting plate **13**. Wherein, the first vertical insertion structure and the first hanging structure are used in combination to limit a vertical downward and left and right lateral movement of the needle grooved plate **21** relative to the mounting plate **13**, ensuring that the second side surface **212** of the needle grooved plate **21** is tightly adhered to the first side surface **131** of the mounting plate **13**, and the first buckling structure is used to limit a vertical upward movement of the needle grooved plate **21** relative to the mounting plate **13**. According to the above embodiment of the present application, the lower portion of the cover plate **22** is detachably combined with the lower portion of the needle grooved plate **21** by a second vertical insertion structure composed of the second limiting feet **223a** and **223b** and the limiting holes **215a** and **215b** on the first limiting feet **214a** and **214b**, the upper portion of the cover plate **22** is detachably hung with the upper portion of the needle grooved plate **21** by a second hanging structure composed of the second hanging holes **224a** and **224b** and the second hooks **217a** and **217b**, and the cover plate **22** is further detachably buckled with the mounting plate **13** by a second buckling structure composed of the top surface **225** of the cover plate **21** and the second elastic buckle **136** on the mounting plate **13**, thereby the cover plate **22** is detachably combined with the needle grooved plate **21**, with the second side surface **222** of the cover plate **22** being in contact with the first side surface **211** of the needle grooved plate **21** to close the plurality of needle guiding grooves **213** to form the plurality of needle guiding holes **23**. Wherein, the second vertical insertion structure and the second hanging structure are used in combination to limit a vertical downward and left and right lateral movement of the cover plate **22** relative to the needle grooved plate **21**, ensuring that the second side surface **222** of the cover plate **22** is tightly adhered to the first side surface **211** of the needle grooved plate **21**, and the second buckling structure is used to limit a vertical upward movement of the cover plate **22** relative to the needle grooved plate **21**. In this way, in the transperineal puncture guiding bracket **100** according to the above embodiment of the present application, not only the guiding assembly **20** and the probe fixing assembly **10** can be detachable and separated, but also the cover plate **22** and the needle grooved plate **21** of the guiding assembly **20** can be detachable and separated, and after being assembled, they can also provide a stable structure to guide the puncture needle **200** to perform precise biopsy.

According to the above embodiment of the present application, the puncture needle **200** can be separated from the transperineal puncture guiding bracket **100**. During a puncture surgery, the puncture needle **200** can be easily separated from the transperineal puncture guiding bracket **100** by operations of pressing the second operating part **1362** of the second elastic buckle **136** at the top of the mounting plate **13** to lift the second buckle position **1361** of the second elastic

buckle **136** and releasing pressure to the cover plate **22**, and then pulling up the cover plate **22** through the handheld members **226a** and **226b** till the second limiting feet **223a** and **223b** at the lower portion of the cover plate **22** is removed from the limiting holes **215a** and **215b** at the lower portion of the needle grooved plate **21**, and then detaching the second hanging holes **224a** and **224b** at the upper portion of the cover plate **22** from the second hooks **217a** and **217b** of the needle grooved plate **21**, and then removing the cover plate **22** laterally.

According to the above embodiment of the present application, the transperineal puncture guiding bracket **100** can be replaced with different needle grooved plates **21** according to puncture needles **200** of different specifications. Firstly, the cover plate **22** can be removed referring to the separation operations as described above. Then the needle grooved plate **21** can be removed by operations of pressing the first operating part **1342** of the first elastic buckle **134** in the middle of the mounting plate **13** to release engagement of the first buckling position **1341** of the first elastic buckle **134** with the limiting buckle **218** of the needle grooved plate **21**, and then pulling up the needle grooved plate **21** till the first limiting feet **214a** and **214b** at the lower portion of the needle grooved plate **21** being removed from the limiting grooves **114a** and **114b** on top of the upper fixing member **11**, and then detaching the first hanging holes **216a** and **216b** at the upper portion of the needle grooved plate **21** from the first hooks **135a** and **135b** of the mounting plate **13**, and then removing the needle grooved plate **21** laterally. Finally, a new needle grooved plate can be installed to replace the needle grooved plate **21** referring to the installation process of the needle grooved plate **21** mentioned earlier.

In the transperineal puncture guiding bracket **100** according to the above embodiment of the present application, the needle grooved plate **21** and the cover plate **22** that constitute the guiding assembly **20** are both made of plastic materials with low cost. According to a second embodiment of the present application, as shown in FIG. 8, a guiding assembly **20'** can be made of metal materials, comprising a metal needle grooved plate **21'** and a metal cover plate **22'**. Both of them can be combined with detachable structures that are basically the same as the needle grooved plate **21** and the cover plate **22** of the aforementioned embodiment to form a plurality of parallel and equidistant metal needle guiding holes **23'**. A biggest advantage of the metal guiding assembly **20'** is that it can prevent the puncture needle from scratching wall of the needle guiding holes **23'** and generating debris.

FIG. 9 and FIG. 10 illustrate exploded structural diagrams of a transperineal puncture guiding bracket **300** according to a third embodiment of the present application respectively. As shown in FIG. 9 and FIG. 10, the transperineal puncture guiding bracket **300** has a similar structure to that of the transperineal puncture guiding bracket **100** in the first embodiment mentioned above, and also comprises a probe fixing assembly **310** and a guiding assembly **320**. Similarly, the probe fixing assembly **310** further comprises an upper fixing member **311**, a lower fixing member **312**, and a mounting plate **313**. The upper fixing member **311** and the lower fixing member **312** work together to hold outer surface of the intracavitary ultrasound probe for fixation, and the mounting plate **313** is provided on the upper fixing member **311** for detachable installation of the guiding assembly **320**. The guiding assembly **320** further comprises a needle grooved plate **321** and a cover plate **322**. The needle grooved plate **321** has a first side surface **3211** and a second side surface **3212** opposite to the first side surface **3211**, and

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the first side surface **3211** of the needle grooved plate **321** is provided with a plurality of vertically equidistant and parallel needle guiding grooves **3213**. The cover plate **322** has a first side surface **3221** and a second side surface **3222** opposite to the first side surface **3221**. The cover plate **322** is also detachably combined with the needle grooved plate **321** to tightly adhere the second side surface **3222** of the cover plate **322** to the first side surface **3211** of the needle grooved plate **321**, thereby closing open sides of the plurality of needle guiding grooves **3213** to form a plurality of needle guiding holes for guiding the puncture needle to perform biopsy puncture.

As shown in FIG. 9 and FIG. 10, there is a through-hole **3133** in the middle of the mounting plate **313** that connects a first side surface **3131** and a second side surface **3132** of the mounting plate **313**, and a first elastic buckle **3134** is set in the through-hole **3133** for clamping the first limiting buckle **3218** in the middle of the second side surface **3212** of the needle grooved plate **21**. The first elastic buckle **3134** has a first buckling position **3135** that extends out of the first side surface **3131** and engages with the first limiting buckle **3218**, and a first operating part **3136** that extends out of the second side surface **3132** for users to operate and release the engagement between the first buckling position **3135** and the first limiting buckle **3218**. The mounting plate **313** is equipped with two first hooks **3137a** and **3137b** on its first side surface **3131**, which extend out from the first side surface **3131** at front and rear ends at an upper portion (preferably at the top) of the first side surface **3131** respectively, for installation with two first hanging holes **3216a** and **3216b** at a corresponding upper portion of the needle grooved plate **321**. There are two limiting grooves **3111a** and **3111b** on top of the upper fixing member **311** adjacent to the mounting plate **313**, which are used for vertical insertion of two first limiting feet **3214a** and **3214b** extending out of the first side surface **3211** at the front and rear ends of the needle grooved plate **321** respectively. Furthermore, there are limiting holes **3215a** and **3215b** on the two first limiting feet **3214a** and **3214b** of the needle grooved plate **321** respectively, for insertion and installation of two second limiting feet **3223a** and **3223b** provided at a corresponding lower portion (preferably at the bottom) of the cover plate **322**. Two second hooks **3217a** and **3217b** are provided at an upper portion of the needle grooved plate **321**, which extend out from the first side surface **3211** at the front and rear ends of the needle grooved plate **321** respectively, for hanging with corresponding second hanging holes **3224a** and **3224b** at the front and rear ends of the cover plate **322**. A difference between the transperineal puncture guiding bracket **300** shown in FIG. 9 and FIG. 10 and the transperineal puncture guiding bracket **100** in the aforementioned first embodiment of the present application is that, a second limiting buckle **3219** is set in the middle of the upper portion of the first side surface **3211** of the needle grooved plate **321**, and a through-hole **3225** is opened in the middle of the upper portion of the cover plate **322**, which passes through the first side surface **3221** and the second side surface **3222** of the cover plate **322**, and a second elastic buckle **3226** is set in the through-hole **3225**, which is used to engage with the second limiting buckle **3219** on the first side surface **3211** of the needle grooved plate **321**. The second elastic buckle **3226** has a second buckling position **3227** that extends out of the second side surface **3222** and engages with the second limiting buckle **3219**, and a second operating part **3228** that extends out of the first side surface **3221** for users to operate and release the engagement between the second buckling position **3227** and the second limiting buckle **3219**. In the

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transperineal puncture guiding bracket **300** according to the third embodiment of the present application, the lower portion of the needle grooved plate **321** is detachably combined with the upper fixing member **311** by a first vertical insertion structure composed of the first limiting feet **3214a** and **3214b** and the limiting grooves **3111a** and **3111b**, the upper portion of the needle grooved plate **321** is detachably hung with the upper portion of the mounting plate **313** by a first hanging structure composed of the first hanging hole **3216a** and **3216b** and the first hooks **3137a** and **3137b**, and the middle portion of the second side surface **3212** of the needle grooved plate **321** is detachably buckled with the middle portion of the first side surface **3131** of the mounting plate **313** by a first buckling structure composed of the first limiting buckle **3218** and the first elastic buckle **3134**, thereby the second side surface **3212** of the needle grooved plate **321** is detachably attached to the first side surface **3131** of the mounting plate **313**. The lower portion of the cover plate **322** is detachably combined with the lower portion of the needle grooved plate **321** by a second vertical insertion structure composed of the second limiting feet **3223a** and **3223b** and the limiting holes **3215a** and **3215b** on the first limiting feet **3214a** and **3214b**, the upper portion of the cover plate **322** is detachably hung with the upper portion of the needle grooved plate **321** by a second hanging structure composed of the second hanging holes **3224a** and **3224b** and the second hooks **3217a** and **3217b**, and the cover plate **322** is further detachably buckled with the first side surface **3211** of the needle grooved plate **321** by a second buckling structure composed of the second elastic buckle **3226** and the second limiting buckle **3219**, thereby the cover plate **322** is detachably combined with the needle grooved plate **321**, with the second side surface **3222** of the cover plate **322** being in contact with the first side surface **3211** of the needle grooved plate **321** to close the plurality of needle guiding grooves **3213** to form the plurality of needle guiding holes. In this way, in the transperineal puncture guiding bracket **300** according to the third embodiment of the present application, not only the guiding assembly **320** and the probe fixing assembly **310** can be detachable and separated, but also the cover plate **322** and the needle grooved plate **321** of the guiding assembly **320** can be detachable and separated, and after being assembled, they can also provide a stable structure to guide the puncture needle to perform precise biopsy.

FIG. 11 shows an exploded structural diagram of a transperineal puncture guiding bracket **400** according to a fourth embodiment of the present application. As shown in FIG. 11, the transperineal puncture guiding bracket **400** has structure that is basically the same as the transperineal puncture guiding bracket **100** in the first embodiment mentioned above, and also comprises a probe fixing assembly **410** and a guiding assembly **420**. Similarly, the probe fixing assembly **410** further comprises an upper fixing member **411**, a lower fixing member **412**, and a mounting plate **413**. The upper fixing member **411** and the lower fixing member **412** work together to hold outer surface of the intracavitary ultrasound probe for fixation. The mounting plate **413** is provided on the upper fixing member **411** for detachable installation of the guiding assembly **420**. The guiding assembly **420** further comprises a needle grooved plate **421** and a cover plate **422**. The needle grooved plate **421** has a first side surface **4211** and a second side surface **4212** opposite to the first side surface **4211**, and the first side surface **4211** of the needle grooved plate **421** is provided with a plurality of vertically equidistant and parallel needle guiding grooves **4213**. The cover plate **422** has a first side surface **4221** and

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a second side surface **4222** opposite to the first side surface **4221**. The cover plate **422** is also detachably combined with the needle grooved plate **421** to tightly adhere the second side surface **4222** of the cover plate **422** to the first side surface **4211** of the needle grooved plate **421**, thereby closing open sides of the plurality of needle guiding grooves **4213** to form a plurality of needle guiding holes for guiding the puncture needle to perform biopsy puncture.

Further as shown in FIG. 11, a through-hole **4133** is formed in the middle of the mounting plate **413** that connects the first side surface **4131** and the second side surface **4132**, and a first elastic buckle **4134** is set in the through-hole **4133** for clamping the first limiting buckle (not visible in the figure) in the middle of the second side surface **4212** of the needle grooved plate **421**. The first elastic buckle **4134** has a first buckling position **4135** that extends out of the first side surface **4131** and engages with the first limiting buckle, and a first operating part that extends out of the second side surface **4132** (not visible in the figure) for users to operate and release the engagement between the first buckling position **4135** and the first limiting buckle. The mounting plate **413** is equipped with two first hooks **4136a** and **4136b** on its first side surface **4131**, which extend out from the first side surface **4131** at front and rear ends at an upper portion (preferably at the top) of the first side surface **4131** respectively, for installation with two first hanging holes **4216a** and **4216b** at a corresponding upper portion of the needle grooved plate **421**. There are two first limiting grooves **4111a** and **4111b** on top of the upper fixing member **411** adjacent to the mounting plate **413**, which are used for vertical insertion of two first limiting feet **4214a** and **4214b** provided at the front and rear ends of the lower portion (preferably at the bottom) of the needle grooved plate **321** respectively. Furthermore, two second hooks **4217a** and **4217b** are provided at an upper portion of the needle grooved plate **421**, which extend out from the first side surface **4211** at the front and rear ends of the needle grooved plate **421** respectively, for hanging with corresponding second hanging holes **4224a** and **4224b** at the front and rear ends of the cover plate **422**. A difference between the transperineal puncture guiding bracket **400** shown in FIG. 11 and the transperineal puncture guiding bracket **100** in the aforementioned first embodiment of the present application is that, there are two second limiting grooves **4112a** and **4112b** on top of the upper fixing member **411** adjacent to the two first limiting grooves **4111a** and **4111b** respectively, for vertical insertion of two second limiting feet (not visible in the figure) at the front and rear ends of the lower portion (preferably at the bottom) of the cover plate **422**. The other difference between the transperineal puncture guiding bracket **400** shown in FIG. 11 and the transperineal puncture guiding bracket **100** in the aforementioned first embodiment of the present application is that, a second elastic buckle **4215** is provided in the middle of the upper portion (preferably the top) of the needle grooved plate **421**. The second elastic buckle **4215** has a second buckling position **4218** extending out of the first side surface **4211** and a second operating part **4219** extending out of the second side surface **4212**. The second buckling position **4216** of the second elastic buckle **4215** presses against the top surface **4225** of the cover plate **422** to form a detachable buckling installation. By operating the second operating part **4219**, buckling by the second buckling position **4216** to the top surface **4225** of the cover plate **322** can be released.

In the transperineal puncture guiding bracket **400** according to the fourth embodiment of the present application, the lower portion of the needle grooved plate **421** is detachably

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combined with the upper fixing member **411** by a first vertical insertion structure composed of the first limiting feet **4214a** and **4214b** and the first limiting grooves **4111a** and **4111b**, the upper portion of the needle grooved plate **421** is detachably hung with the upper portion of the mounting plate **413** by a first hanging structure composed of the first hanging hole **4216a** and **4216b** and the first hooks **4136a** and **4136b**, and the middle portion of the second side surface **4212** of the needle grooved plate **421** is detachably buckled with the middle portion of the first side surface **4131** of the mounting plate **413** by a first buckling structure composed of the first limiting buckle (not visible in the figure) and the first elastic buckle **4134**, thereby the second side surface **4212** of the needle grooved plate **421** is detachably attached to the first side surface **4131** of the mounting plate **413**. The lower portion of the cover plate **422** is detachably combined with the lower portion of the needle grooved plate **421** by a second vertical insertion structure composed of the second limiting feet (not visible in the figure) and the second limiting grooves **4112a** and **4112b**, the upper portion of the cover plate **422** is detachably hung with the upper portion of the needle grooved plate **421** by a second hanging structure composed of the second hanging holes **4224a** and **4224b** and the second hooks **4217a** and **4217b**, and the upper portion of the cover plate **422** is further detachably buckled with the needle grooved plate **421** by a second buckling structure composed of the top surface **4225** of the cover plate **422** and the second elastic buckle **4215** of the needle grooved plate **421**, thereby the cover plate **422** is detachably combined with the needle grooved plate **421**, with the second side surface **4222** of the cover plate **422** being in contact with the first side surface **4211** of the needle grooved plate **421** to close the plurality of needle guiding grooves **4213** to form the plurality of needle guiding holes. In this way, in the transperineal puncture guiding bracket **400** according to the fourth embodiment of the present application, not only the guiding assembly **420** and the probe fixing assembly **410** can be detachable and separated, but also the cover plate **422** and the needle grooved plate **421** of the guiding assembly **420** can be detachable and separated, and after being assembled, they can also provide a stable structure to guide the puncture needle to perform precise biopsy.

The above is only the preferred embodiments of the present application, and is not intended to limit the present application. Any modifications, equivalent substitutions and improvements made within the spirit and principles of the present application should be included within the protection scope of the present application.

We claim:

1. A transperineal puncture guiding bracket, comprising a probe fixing assembly and a guiding assembly, wherein the probe fixing assembly is fixedly installed with an intracavitary ultrasound probe, the guiding assembly provides a plurality of vertically distributed and parallel needle guiding holes along a puncture direction, characterized in that, a mounting plate is set on top of the probe fixing assembly, and the guiding assembly further comprises a needle grooved plate and a cover plate, the needle grooved plate has a first side surface and a second side surface opposite to the first side surface, with the first side surface of the needle grooved plate being provided with a plurality of vertically distributed and parallel needle guiding grooves; and a lower portion of the needle grooved plate is detachably combined with the top of the probe fixing assembly adjacent to the mounting plate by a first vertical insertion structure, an upper portion of the needle grooved plate is detachably

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hung with an upper portion of the mounting plate by a first hanging structure, and the second side surface of the needle grooved plate is detachably buckled with a first side surface of the mounting plate by a first buckling structure, thereby the second side surface of the needle grooved plate is detachably attached to the first side surface of the mounting plate; and a lower portion of the cover plate is detachably combined with the lower portion of the needle grooved plate or the top of the probe fixing assembly by a second vertical insertion structure, an upper portion of the cover plate is detachably hung with the upper portion of the needle grooved plate by a second hanging structure, and the cover plate is further detachably buckled with the upper portion of the mounting plate, the upper portion of the needle grooved plate or the top of the probe fixing assembly through a second buckling structure, thereby a second side surface of the cover plate is detachably attached to the first side surface of the needle grooved plate to close the plurality of needle guiding grooves to form the plurality of needle guiding holes.

2. The transperineal puncture guiding bracket of claim 1, characterized in that the first vertical insertion structure comprises first limiting feet provided at the lower portion of the needle grooved plate which extend out from the first side surface of the needle grooved plate and limiting grooves formed at the top of the probe fixing assembly for insertion of the first limiting feet; and the second vertical insertion structure comprises second limiting feet set at the lower portion of the cover plate and limiting holes on the first limiting feet of the needle grooved plate for insertion of the second limiting feet.

3. The transperineal puncture guiding bracket of claim 1, characterized in that the first vertical insertion structure comprises first limiting feet provided at the lower portion of the needle grooved plate and first limiting grooves set at the top of the probe fixing assembly for insertion of the first limiting feet; and the second vertical insertion structure comprises second limiting feet set at the lower portion of the cover plate and second limiting grooves formed at the top of the probe fixing assembly for insertion of the second limiting feet.

4. The transperineal puncture guiding bracket of claim 1, characterized in that the first hanging structure comprises first hanging holes at the upper portion of the needle grooved plate and first hooks at the upper portion of the mounting plate which extend out from the first side surface of the mounting plate to hang into the first hanging holes; and the second hanging structure comprises second hanging holes at the upper portion of the cover plate and second hooks at the upper portion which extend out from the first side surface of the needle grooved plate to hang into the second hanging holes.

5. The transperineal puncture guiding bracket of claim 1, characterized in that the first buckling structure comprises a first limiting buckle set in the middle of the second side surface of the needle grooved plate and a first elastic buckle set in the middle of the mounting plate, wherein a through-

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hole is set in the middle of the mounting plate that connects the first side surface and a second side surface of the mounting plate, and the first elastic buckle is set inside the through-hole, the first elastic buckle has a first buckling position that is engaged with the first limiting buckle and a first operating part that extends out of the second side surface of the mounting plate for releasing engagement between the first buckling position and the first limiting buckle.

6. The transperineal puncture guiding bracket of claim 1, characterized in that the second buckling structure comprises a second elastic buckle set at the upper portion of the needle grooved plate, wherein the second elastic buckle has a second buckling position for pressing a top surface of the cover plate and a second operating part for releasing the second buckling position from pressing the top surface of the cover plate.

7. The transperineal puncture guiding bracket of claim 1, characterized in that the second buckling structure comprises a second elastic buckle set at the upper portion of the mounting plate, wherein the second elastic buckle has a second buckling position for pressing a top surface of the cover plate and a second operating part for releasing the second buckling position from pressing the top surface of the cover plate.

8. The transperineal puncture guiding bracket of claim 1, characterized in that the second buckling structure comprises a second limiting buckle set at the upper portion of the first side surface of the needle grooved plate and a second elastic buckle at the corresponding upper portion of the cover plate, wherein a through-hole is set at the upper portion of the cover plate that connects a first side surface and the second side surface of the cover plate, and the second elastic buckle is set inside the through-hole, the second elastic buckle has a second buckling position that is engaged with the second limiting buckle and a second operating part that extends out of the first side surface of the cover plate for releasing engagement between the second buckling position and the second limiting buckle.

9. The transperineal puncture guiding bracket of claim 1, characterized in that the probe fixing assembly comprises an upper fixing member and a lower fixing member, wherein the upper fixing member and the lower fixing member work together to hold outer surface of the intracavitary ultrasound probe, and the mounting plate is set on top of the upper fixing member.

10. The transperineal puncture guiding bracket of claim 9, characterized in that upper wrapping surfaces are formed on a lower surface of the upper fixing member for fitting upper surface of the intracavitary ultrasound probe, and lower wrapping surfaces are formed on an upper surface of the lower fixing member for fitting lower surface of the intracavitary ultrasound probe, and the upper fixing member and the lower fixing member are rotationally connected on one side and buckled on the opposite side to tightly hold the intracavitary ultrasound probe through the upper wrapping surfaces and the lower wrapping surfaces.

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